

IMC Prosperity 4 Manual Challenge

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May 2026

1 Introduction

IMC Prosperity is a quant trading competition that runs for several weeks each spring. This year, I participated for the first time in a group of 5 against a competitor pool of 18,800+ teams globally. There are several rounds; in each, there is a manual and an algorithmic component. The algo component involves constructing and implementing a strategy for some theoretical market, provided some amount of historical data. The manual component is usually more puzzle-like, involving optimization, game theory, and market microstructure knowledge. My responsibility this year was completing the manual challenges, as they align more closely with my expertise than the pure Python implementation of the algo side of things.

I hope to compete again next year, and expand my contributions past the manual challenges.

2 Round 1: “Trading Groundwork”

As the first manual challenge of the competition, this round was relatively straightforward and entirely deterministic. The rules can be found below.

2.1 Round Outline

Provided two markets, one for Dryland Flax and another for Ember Mushrooms, we are asked to submit a single limit order (price, quantity) in each. This particular exchange operates with a unique structure - when the auction ends, the exchange selects a single clearing price that:

1. maximizes total traded volume, then
2. breaks ties by choosing the higher price.

Once a clearing price is selected, all bids with prices above the clearing price (inclusive) are filled. Allocation is price priority, then time priority. In this puzzle, we are the last to submit orders to the market, so although we benefit from complete information, we are the last in line at any price level we join.

Immediately after the auction concludes and orders are filled, any inventory we buy is bought back at a fixed price (\$30 per unit of flax with no fees, and \$20 per unit of mushrooms with a \$0.10 fee per unit traded).

2.2 Approach

I noticed that the optimal solution is found at some “edge” at which an incremental volume adjustment would alter the clearing price of the order book. By positioning our order volume right beneath that threshold, we could improve the price we buy flax at, with a marginal volume hit.

For example, bidding for 5,000 flax at \$29 forced a clearing price of \$29, and thus a total profit of \$5,000. But bidding for 4,999 flax at the same price forced a clearing price of \$28, and thus a profit of \$9,998.

I generalized this approach by reasoning that the optimal bid quantity at a given price will be found at a clearing price “transition” point. Using this logic, I Cursor-ed a quick Python script to find the optimal bid strategy in the flax market. I then tweaked the script for the mushroom market by adding in the fee structure. The optimal bid ended up being 19,999 mushrooms at \$17 for a profit of \$78,000 (for reference, adding one mushroom to our bid would have reduced profit by \$20,000).

Result: 2nd place globally in Round 1 manual challenge.

Reflection: A number of other teams found the same solution as us, and several teams were able to generate \$1 additional in profit. This bugged me (and still does), but since I lost access to the round 1 challenge portal after round 2 began, I may never know for certain. I hypothesize that either there was a solution past the bounds I restricted my script to (I only let it check values I found intuitively “reasonable” as I was being very impatient) or there was some strategy in which submitting an ask instead of a bid was slightly better (I didn’t check this logic, either, as upon first glance it did not appear to be profitable). However in the grand scheme of things, a \$1 price difference from 1st place had no say in the overall results of the competition.

3 Round 2: “Growing Your Outpost”

This was the final round of the qualifiers (where each team had a \$200,000 total PnL goal to meet in order to qualify for the finals). This round deviated from the purely deterministic nature of round 1, which I found to be highly enjoyable. The rules can be found below.

3.1 Round Outline

Each team is given a \$50,000 budget to allocate across 3 “pillars”:

1. Research

2. Scale

3. Speed

We choose percentage values of our total budget to allocate to each pillar. Final PnL is calculated as

$$\text{PnL} = (\text{Research} \times \text{Scale} \times \text{Speed}) - \text{Budget Used}$$

However, each pillar handles budget allocation differently. Research grows logarithmically from 0 (for 0% invested) to 200,000 (for 100% invested) according to the formula

$$\text{research}(x) = 200,000 \times \frac{\ln(1+x)}{\ln(1+100)}$$

for budget allocation percentage $x \in \mathbb{Z}$. Scale grows linearly from 0 (for 0% invested) to 200,000 (for 100% invested). We can write this as

$$\text{scale}(x) = 200,000 \times \frac{x}{100}$$

Finally, speed is determined by how your budget allocation compares to the other teams'. The highest speed investment receives a 0.9 multiplier, the lowest investment receives a 0.1 multiplier, and everyone in between is scaled linearly by rank.

3.2 Approach

Before even touching the speed budget allocation, we can optimize our allocation to the other two pillars if we treat speed as a constant k . With allocations r = research, s = scale, and v = speed, we can write PnL as:

$$\begin{aligned} \text{PnL} &= \left(\frac{7s}{100} \right) \left(200,000 \cdot \frac{\ln(1+r)}{\ln(101)} \right) (k) - 50,000 \\ &= \left(\frac{7k}{2,000 \cdot \ln(101)} \right) (s \cdot \ln(1+r)) - 50,000 \end{aligned}$$

Thus we maximize PnL at the maximum of $f(r, s) = s \cdot \ln(1+r)$. Using our restriction $100 = r + s + v$, we get the equations

$$v = 100 - r - (1+r) \cdot \ln(1+r)$$

$$s = 100 - v - r$$

My Numerical Analysis class had just covered Newton's fixed point iterations, and this seemed like a perfect place to try it out. I used Newton's method with $f(r) = v - 100 + r + (1+r) \cdot \ln(1+r) = 0$ and 4 iterations. Given x_1 as the current guess for r and x_2 as the updated guess,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

I wrote a Python script to output our optimized research & scale allocations when provided a speed allocation value (mostly reused a program from a homework set where I had to do something similar). Then I prompted Cursor to write a script to optimize our speed allocation given a distribution of opponent speed allocations. I ran this script 100 times each on normal, left-skewed normal, log-normal, geometric, and uniform distributions (each of which I fit to the constraints of this problem) to see an aggregate “best” strategy.

Distribution	Aggregate Best Speed Allocation	PnL
Geometric	22.06	\$167,741.73
Log-Normal	40.13	\$250,445.37
Left-Skewed Normal	50.80	\$208,635.86
Normal	47.24	\$164,537.52
Uniform	37.50	\$149,849.59

The best speed allocation values varied pretty heavily (between 22% and 50%), so I knew I needed more of an assumption regarding opponent behavior to eliminate as much luck as possible. I ended up deciding to model competitor behavior with a log-normal distribution ($\mu = 3.5, \sigma = 0.5$) to account for a relatively high density of risk-averse players and a long tail of hyper-rational/greedy exceptions. I rounded up to a 41% budget allocation to speed (and thus 15% to research and 44% to scale) to position ourselves just past the point of highest competitor density.

Result: 11th place globally in round 2 manual challenge.

Reflection: I was extremely proud of this placement - if I remember correctly, our competition ended up placing themselves such that a 38% speed allocation was optimal, with a fairly log-normal distribution. I don’t think any other team had a fundamentally “better” approach than we did.

4 Round 3: “Gloves Off”

This was the first round of the finals. Keeping with the game theory theme of round 2, this round required us to make some form assumption of our opponents’ behaviors to win. As always, the rules for this round can be found below!

4.1 Round Outline

This round asks us to submit two bids for a product in a competitive market. Our first bid is compared to “reserve prices” held by counterparties (which are uniformly distributed on $[670, 920]$ in increments of 5). If our bid is higher than a reserve price, we trade at our bid.

Our second bid is compared to reserve prices AND the average of all second bids among all other players. If our bid is higher than the mean, then we

trade with the same execution as above (trading at our bid price with every counterparty with a reserve price less than or equal to our bid). If our bid is lower than the mean, however, our PnL is penalized by the following multiplier:

$$\left(\frac{920 - \text{avg}_{b_2}}{920 - b_2} \right)^3$$

Clearly, we want to avoid this penalty. Focusing on the first bid, we can write our PnL function as

$$\text{PnL} = \text{margin} \times \text{volume}$$

Writing margin as $(920 - \text{bid})$ and volume as $\lfloor \frac{x-670}{5} \rfloor$, we get for bid price x a total PnL of

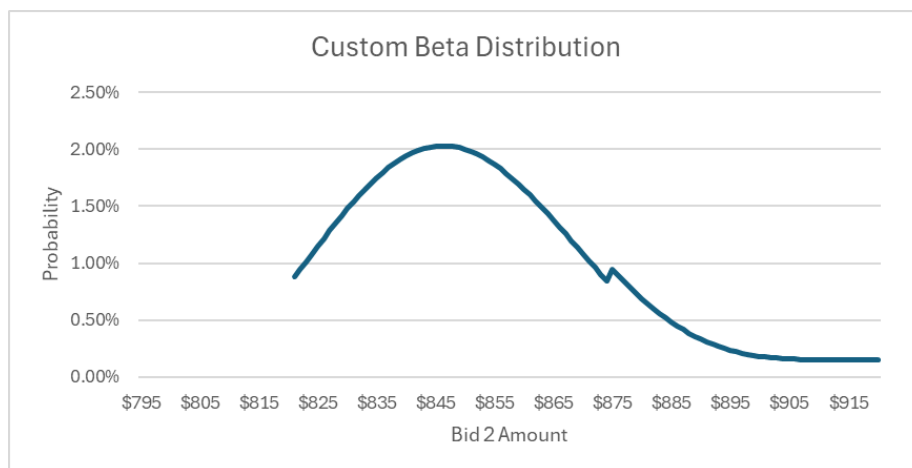
$$\text{PnL} = \left\lfloor \frac{x - 670}{5} \right\rfloor \times (920 - x)$$

Thus for isolation bid 1, PnL is optimized with a bid of \$795. In other words, if we expect \$0 PnL from bid 2, the highest our bid 1 should go is \$795. But the more confident we are in transacting with bid 2 (e.g. relatively low avg_{b_2} , giving good margins), our bid 1 should be lowered to allow our B2 to capture more volume, while simultaneously transacting with our bid 1 at more favorable margins. I wrote a quick script to calculate PnL for every B1/B2/avgB2 combination, and got the following results: Notice that an optimal bid 2 is \$1

Average B2	Optimal B1	Optimal B2
\$840	\$756	\$841
\$845	\$761	\$846
\$850	\$761	\$851
\$855	\$766	\$856
\$860	\$766	\$861
\$865	\$771	\$866
\$870	\$771	\$871
\$875	\$776	\$876
\$880	\$776	\$881
\$885	\$781	\$886
\$890	\$781	\$891
\$895	\$786	\$896
\$900	\$786	\$901
\$905	\$791	\$906
\$910	\$791	\$911

above the average - since we transact when bid i clearing price (with strict inequality), we are able to capture additional volume at a relatively small margin reduction. Now by attempting to model our opponents' behavior, we can try and get a decent guess for our own behavior. I chose a beta distribution (my favorite) with $\alpha = 5, \beta = 6.25$ to set the mode opponent guess around the "alleged" Nash Equilibrium of the game at $B_2 = 836$ (I didn't even bother doing this or verifying; it was the agreed upon value in the competition's discord

channel, and I knew that enough people would deviate that operating purely “rationally” was irrational in this case). I then manually detached the density below the \$821 mark and relocated it to the right tail, in an attempt to account for a fair proportion of overly rational/greedy competitors bidding around the \$900 mark:



This is obviously not entirely accurate, but I think it’s a pretty decent heuristic. This distribution resulted in an expected mean of \$854.81. My group felt pretty confident about their algorithm for this round, so I played it safe and bid \$875.

Result: 165th globally in round 3 manual.

Reflection: This was my first huge mistake of the competition. I got nervous; I didn’t want to bid below the average and tank our teams’ standing. If I had followed the math, we would have bid \$860. The average ended up being \$859.

This was definitely a good learning experience - I designed a mathematical framework to model my uncertainty in opponent behavior. In this framework I accounted for wanting to “play it safe” by artificially implanting volume on the aggressive bidder long tail. Playing it even safer than that was irrational, and it ended up costing us.

However, to be fair to myself as hindsight is indeed 20/20, my math was not perfect. The average was slightly higher than anticipated, and I would have come very close to subjecting us to the underbidding penalty. I think there’s probably a middle ground that would have been a little bit more justifiable, but even with our decision we still placed fairly well (there were thousands of teams remaining at this point in the competition).

5 Rounds 4 and 5

Unfortunately, this is where the bulk of my participation in this competition came to an end. In the finals, rounds are spaced pretty close together, and I had an unfortunately timed trip to Washington, D.C. to take a polygraph and participate in other, similarly unpleasant, security clearance processing-related tasks.

Round 4 involved trading “Aether Crystal” vanilla options with 2 and 3 week expiries, and several exotic derivatives written on the same underlying. I regretted missing this one - I don’t know a ton about derivatives, and I think this would have been a fascinating way to learn a little.

Round 5 was less interesting to me - I was able to work on it a little bit on the plane. It was highly discretionary and strayed away from the puzzle-ness of the previous rounds. We were provided several news articles regarding hypothetical companies and their products (surrounding things like macro conditions and factory shutdowns). We were supposed to use this information to form opinions on the movement of the stocks during the next trading day. The one interesting catch was that the more we traded in one good, the more expensive the good got. We ended up grounding our portfolio in our own opinions, and using previous years’ info to “check” our work. Since the final round was the same last year (with different companies / news articles), our team matched similar companies / articles and observed how IMC decided to move their stock values. We ended up placing fine relative to our opponents, but not exceptionally well. I remember being somewhat frustrated with the lack of numerical information we could find, but I can’t really complain given everyone had access to the same info.

6 Conclusion

All in all, we placed **71st out of 18,803 teams globally, and 22nd in the USA** (out of an unknown-to-me number of teams). This feels very strong to me, but I have to admit I was hoping for a stronger final placement, given we sat at 4th globally after the first round of finals, and 9th globally after the second. But I definitely can’t complain - I wouldn’t have gotten anywhere close to this without the whole team.

In short, I learned a ton, and I’m very happy I participated.